MILITARY SPECIFICATION

MACHINING CENTERS, VERTICAL, SINGLE SPINDLE, DOUBLE COLUMN, COMPUTER NUMERICALLY CONTROLLED (CNC)

The specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers computer numerically controlled (CNC) vertical spindle, double column machining centers with equipment and accessories specified herein.

1.2 Classification. This specification covers machining centers of the following types and sizes. The type and size to be furnished shall be as specified (see 6.2.1).

Type I - Traveling cross rail

	Minimum spindle motor horsepower	Table size, inches width by length (optional length)
Size 1	20	38 by 60
Size 2	20	38 by 70 (110)
Size 3	20	45 by 80
Size 4	25	50 by 90 (120) (180)
Size 5	30	78 by 140 (180)
Size 6	30	90 by 180

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3408
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Type II - Fixed cross rail

	Minimum spindle motor horsepower	Table size, inches width by length (optional length)
Size 1	10	30 by 50
Size 2	25	38 by 110 (140)
Size 3	25	41 by 64
Size 4	25	42 by 98
Size 5	25	48 by 32
Size 6	25	50 by 140 (180)
Size 7	25	60 by 146
Size 8	25	70 by 140 (180)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues,

ANSI B5.50 - "V" Flange Tool Shanks for Machining Centers with Automatic Tool Changers.

ANSI B11.8 - Drilling, Milling, and Boring Machines - Safety Requirements for the Construction, Care and Use.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-232 - Interface between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange.

RS-244 - Character Code for Numerical Machine Tool Control Perforated Tape.

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machine Tools.

RS-274 - Interchangeable Variable Block Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines.

RS-281 - Construction Standards - Numerical Machine Tool Control.

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RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape.

RS-494 - 32 Bit Binary CL Exchange (BCL) Input Format for Numerically Controlled Machines.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 20006.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA '22209-1695.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text Of this specification shall take precedence. Nothing in this specification,

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however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design. The machining center shall be new and one of the manufacturer's current models. Its design shall include features and components necessary for maintaining alignment and accomplishing drilling, tapping, milling, and boring as required herein. The machine shall have a vertical axis spindle assembly mounted on the spindle carriage. Z-axis motion shall be provided by vertical movement of the spindle assembly or quill. The spindle carriage shall move along the guideways of the cross rail; this is Y-axis movement. On type I machines, the cross rail moves in a vertical direction along guideways of the columns, W-axis movement. On type 11 machines, the cross rail is rigidly mounted to the two columns. The columns shall extend down either side of the worktable and attach to the machine base. The base shall support a worktable arranged to support and secure the workpiece in the machining operations. The table shall move longitudinally along ways of the bed for X-axis motion. Tools, toolholders, and adapters shall be inserted and removed from the spindle by a tool changer with storage magazine. Tool change, spindle operation, and motion in the X, Y, and Z axes shall be directed by a computer numerical control (CNC) system. The machine and its numerical control system shall constitute a completely functional system with built-in operations programmed by coded punch tape or other devices. The machine functions shall be controlled by manually operated control devices, semi-automatically by manual data input devices, and automatically from tape input and part programs in data memory. Al 1 parts of the machine or system that are subject to wear, breakage, or distortion shall be accessible for adjustment, replacement and repair.

3.2 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units CUS) or the International System of Units (SI) shall be used in the design and construction of the machining center. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices, When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1), Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

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3.2.2 Reclaimed materials. The machining center may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machining center and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of 29 CFR 1910 that are applicable to the machine itself. In addition, the machine shall comply with all requirements of ANSI B11.8 that are designated therein as the responsibility of the machine manufacturer. In the event of conflict between the requirements of OSHA and the ANSI standards, the requirements of OSHA shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at point of installation.

3.2.9 Lubrication. An automatic lubricating system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by oil level sight gauge, a low oil level warning light, or a similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

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3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machining center shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. No processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent loosening. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer’s commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machining center shall be painted in accordance with the manufacturer’s commercial practice, provided that all surfaces to be painted shall be cleaned of all foreign matter and at least one coat of primer and one coat of finish color is applied.

3.3.6 Threads. All threaded parts used on the machining center and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears and pinions spindle and axis drive train shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

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3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machining center shall include components consisting of a base, columns, cross rail, worktable, machining head, spindle assembly, automatic tool changer, coolant system, electrical system, and numerical control system.

3.4.1 Base. The base shall be a casting of iron or iron alloy or a fabricated steel weldment, ribbed and braced to minimize distortion and deflection. The base shall form a foundation for the entire machine, with the columns machined and securely attached to provide sufficient rigidity and mutual component alignment for assuring the accuracy required herein. The base shall have guideways that are proportioned to fully support the worktable throughout its full range of travel. The base shall have means for leveling the machine on a flat mounting surface. Hydraulic or coolant fluid reservoirs may be housed within the base or be supplied as separate external units. Reservoirs shall have means for indicating fluid levels. The base shall have access doors or removable plates as necessary for servicing components housed within.

3.4.2 Columns. The columns shall be castings of iron or iron alloy or fabricated steel weldments of box-like construction, internally ribbed and braced to minimize distortion and deflection under normal tool loads. The columns shall be designed to support and accommodate the cross rail and, for type I machines, have guideways to support the vertical travel of the cross rail. On type 11 machines, the cross rail shall be rigidly mounted to the columns.

3.4.3 Worktable. The worktable shall be a casting of iron or iron alloy, internally ribbed and proportioned to support and withstand maximum capacity workpiece weights and tool thrust loads. The worktable shall be precision fitted to the ways of the base and be arranged to provided workpiece travel in the X-axis. The top of the worktable shall be machine finished for a work mounting surface. The worktable shall have a coolant trough or similar means for draining spent coolant into its reservoir. Unless otherwise specified (see 6.2.1), the number of the T-slots in the worktable top shall be the manufacturer’s standard. The size and configuration of the T-slots shall be in accordance with ANSI B5.1. When required, the worktable shall be furnished with a sub-plate of the characteristics specified (see 6.2.1).

3.4.4 Spindle carriage. The spindle carriage shall be a casting of iron or iron alloy or a fabricated steel weldment designed and proportioned to house and support the quill (as required), spindle assembly, and spindle rotational drive with motor and related components necessary for accomplishing drilling, tapping, milling, and boring required herein. The spindle carriage shall hold the spindle in axial alignment perpendicular to the work

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mounting surface of the worktable and shall be designed to prevent distortion to the head due to torsional forces of the drives. The machine shall have stops that are located at the upper and lower extremes of Z axis travel to prevent excessive overtravel.

3.4.4.1 Spindle assembly. The spindle shall be of alloy steel hardened and ground to a finish not rougher than 32 microinch. The spindle shall be supported and held in axial and radial alignment by ball or roller bearings. The spindle shall be mounted in the machining head with its bearings designed and arranged to compensate for thermal expansion. The spindle shall have positive means for accepting, retaining and releasing tools and tool holders inserted by either the automatic tool changer or the operator. Unless otherwise specified (see 6.2.1), the spindle shall accept sizes 24 or 50 “V” flange tool shanks complying with ANSI B5.50. The spindle shall be equipped to orient the automatically inserted tools to maintain the same radial position they occupied in the tool changer. The machine shall drive the spindle in both the clockwise and counterclockwise directions of rotation for accomplishing the full range of spindle speeds specified herein. The machine shall be arranged to automatically stop spindle rotation for tool change. The machine shall have overload protection which automatically stops the movement of all slides and the rotation of the spindle should it stall from overload.

3.4.5 Tool changer. The machine shall have an automatic tool changer with a magazine capable of storing not less than the tool capacity required for the style machine specified in the contract or order, The magazine shall be of the rotary drum design or of any other closed loop recirculating configuration with the capability of presenting a preprogrammed tool to the tool changer for exchange with the tool held in the spindle at the time of programmed commands for tool change. Tool change shall be accomplished by means of a power-actuated mechanism which shall either interchange the tool in the spindle with a tool selected from the tool magazine or remove the tool from the spindle and then insert the tool from the magazine. The magazine shall permit convenient loading and unloading of tools, tool holders, and adapters during operational set-up. The tool changer shall be arranged for random tool selection which shall include the capability of reusing a tool that has been returned to the magazine after a program operation.

3.4.6 Guideways. All guideways shall be designed and constructed to permit precision control and movement of mating components. The guideways shall be hardened and ground. Mating slides shall have means of maintaining alignment to prevent cutting forces from separating slide surfaces or distorting feed screws. The guideways shall be shielded from chips, coolant fluid and other contaminants by telescoping metal covers or similar means.

3.4.7 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a

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baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.8 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Over pressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.9 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.9.1 Motors and motor accessories. Motors shall have ball or roller bearings and shall be rated for continuous duty. Motors shall be either alternating-current (AC) 3-phase, 60-Hz, or industrial direct-current (DC) motors conforming to ANSI/NEMA Standard MG 1 and equipped with solid state rectifiers, or shall be stepping type motors with not less than 50 incremental steps per revolution. Unless otherwise specified (see 6.2.1), each electric motor shall meet the requirements of a dripproof enclosure.

3.4.10 Numerical control. The machining center shall have a full automatic solid state softwired, integrated circuit type numerical control. The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, three linear axes movement, spindle operation, tool change, and other part program directed functions for machining centers of the type specified herein. Controlled axes shall be identified in accordance with EIA Standard RS-267. The control shall have linear and circular interpolation and be capable of simultaneous control of three linear axes. Control features shall include a programmable interface, buffer storage, fixed cycles, part program storage, part program and buffer edit capability, and control diagnostics. The MCU shall direct machine functions from part program data stored in memory, manually input, or input directly from the media specified herein. The control shall initiate a halt or an error signal should a fault condition occur in the control. All necessary executive program routines shall be furnished in the form of 1-inch eight-channel binary coded decimal punched tape or flexible diskette for controls having volatile memory. The control unit shall conform to EIA Standard RS-281. A line voltage of +10 percent from normal shall not adversely affect the numerical control system function. The control shall be capable of functioning in ambient temperatures ranging from 30 to 120″ Fahrenheit

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and humidity ranging from 5 to 95 percent noncondensing. The control shall automatically shut down when internal operating temperature exceeds safe operating temperature. Control resolution shall be no greater than 0.0001 for each linear axis and not more than 0.001 of a degree for each rotary axis. When binary cutter location (BCL) data is specified (see 6.2.1), the numerical control shall have the necessary software to perform all geometry dependent functions which are ordinarily performed by the postprocessor.

3.4. 10.1 MCU. The MCU shall provide full contouring control of the machine's three linear axes for performing drilling, tapping, boring, reaming, and contour milling operations by simultaneous movement of each axis.

3.4.10.2 Data input.

3.4.10.2.1 Punched tape data input. Unless otherwise specified (see 6.2.1), the input media for the control shall be l-inch, eight-channel punched tape. The data input format and codes shall comply with EIA Standards RS-274 and RS-358 or, when specified (see 6.2.1), RS-244 in lieu of RS-358. The control shall be equipped with a photoelectric tape reader capable of holding, feeding and reading tape prepared in the format specified at a rate not less than 150 characters per second. Unless otherwise specified (see 6.2.1), the tape reader shall be supplied with standard feed and take-up spools 5-1/4 inches in diameter. The tape reader shall be housed in a dust free area of the numerical control cabinet and shall be easily accessible for cleaning and tape handling. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA RS-244.
b. Even parity per EIA RS-358.
c. Misaligned sprocket holes.
d. No sprocket holes.

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.4.1O.2.2 Flexible diskette data input. When 32 Bit Binary Cutter Location (BCL) Exchange Input for Numerically Controlled Machine is specified (see 6.2.1), the input media shall be an 8-inch flexible diskette recorded in accordance with 3.4.10.2,2.1. The input data format shall comply with EIA Standard RS-494. Functions 10.1, 10.2 and 10.3 of section 10, EIA RS-494 shall apply. Manufacturers providing functions or features that cannot be addressed by command codes currently covered by EIA RS-494 shall apply to the Electronic Industries Association, IE-31 Committee on Numerical Control Systems and Equipment for an extension of the standard codes to cover the new features or functions. The control shall be equipped with an 8-inch flexible diskette reader. The reader shall have the capability to perform a cyclic redundancy check of the input data and be capable for reading not less than 15,000 characters per second.

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3.4.10.2.2.1 Flexible diskette format. When an 8-inch flexible disk is specified as the input media to the numerical control it shall be a single density recording on one side using a soft-sectored format. The disk shall be divided into 77 distinct locations on the single side disk. Tracks shall be assigned numbers from 00 through 76. Track 00 shall be nearest to the edge of the disk and track 76 nearest the center. Tracks 00, 74, 75 and 76 shall not be used. There is an index hole in the jacket and the disk which shall be used for timing functions when a beam of light passes through it. Recording shall be accomplished by rotating the disk inside its jacket exposing the disk at the read/write head slot cut in the jacket. Data shall be recorded only on the side of the disk that is exposed to the read/write head. Each track shall be divided into 26 addressable sectors which contain 128 bits. All sectors shall be identified by the file starting sector number with track 00, sector 1, designated as 0. The first part program shall be located at sector 1E hexadecimal corresponding to track 1, sector 5. Where two byte numbers are referenced the high order byte shall be recorded first and the low order last. The data stored in a sector is called a record, Track 1, sectors 1 through 4 shall contain the program directory of up to 16 part program names in ASCII. Each file shall be packed on the disk in the same order as its directory entry, Unused directory entries shall point to the next available sector after the last file and contain the 30-character ASCII space characters. The last sector of each file shall be padded with end-of-file (EOF) records (8000 0000). Diskette addressing shall comply with the requirements of table I.

3.4.10.2.3 Manual data input. The MCU shall have a manual data input (MDI) feature to permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.10.3 Data display. A cathode ray tube (CRT) capable of displaying at least 250 characters simultaneously, shall be provided. Unless otherwise specified (see 6.2.1), the display shall, as a minimum, display the following data:

a. Four digit block sequence number.
b. Active block of stored data.
c* Actual position of each axis.
d. Feedrate and override condition.
e. Control and system error messages.

3.4.10.4 Control system modes of operation. The operational modes of the CNC control shall include manual, automatic, single block, MDI, sequence number search, and optional stop.

3.4.10.5 Departure control. The maximum departure permitted by a single command in either incremental or absolute mode shall be not less than plus or minus 99.9999 inches. The control shall be capable of automatically

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controlling the acceleration and deceleration of the slides. The resolution of slide movement in either direction shall not be more than 0.0001 inch or 0.001 mm.

3.4.10.6 Absolute/incremental input. The control shall have absolute/ incremental input capability. This mode shall be selected by a preparatory G-code which may be switched during the tape program. (Not required if BCL input is specified (see 6.2.1).

3.4. 10.7 Automatic tape code recognition. The control shall have the capability to automatically recognize and read tape punched with either EIA RS-244 or RS-358 (ASCII) character codes. (Not required if BCL data input is specified (see 6.2.1).

3.4.10.8 Switchable inch/metric input. The control system shall provide switchable inch/metric input capability.

3.4.10.9 Part program storage. Unless otherwise specified (see 6.2.1), part program storage shall be in either solid-state or bubble-type memory. The MCU shall be capable of storing in memory the content of not less than 250 feet of postprocessed part program tape data unless otherwise specified (see 6.2.1), (Not applicable if BCL data input is specified (see 6.2.1).)

3.4.10.10 Standby battery power. Unless otherwise specified (see 6.2.1), standby battery power shall be provided for controls having volatile memory, to hold the memory provided at full load for a period of not less than 72 hours when the primary power source fails.

3.4.10.11 Part program edit. A program edit capability shall be provided to allow block insertion, deletion or modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.10.12 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from zero to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that programmed.

3.4.10.13 Spindle speed override. A spindle speed override shall be provided to allow manual adjustment of spindle speed in either continuous adjustment or in steps of not greater than 10 percent each. The spindle speed override control shall control not less than the range of override percentages between 60 percent and 120 percent of programmed speed. The spindle speed override control shall have a position that allows return of speed to that programmed.

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3.4.10.14 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequence number.

3.4.10.15 Mirror image. The machine shall be capable of producing either right- or left-hand parts from the same tape data through the inversion of X and Y axes. Inversion of each axis shall be by keyboard or switch entry. Mirror image shall not affect the directional sense of manual controls. Reverse settings shall be displayed on the data display.

3.4.10.16 Preparatory function (G-Codes). The preparatory function codes of the control shall be in accordance with EIA Standard RS-274. If BCL data input is specified, codes in EIA-RS-494 shall apply.

3.4.10.17 Miscellaneous functions (M-Codes). The miscellaneous function codes of the control shall be in accordance with EIA Standard RS-274. If BCL data input is specified, codes in EIA RS-494 shall apply.

3.4.10.18 Block delete. The control shall have block delete feature to permit by-passing of blocks of programmed data.

3.4.10.19 Buffer storage. The control shall have buffer storage for transferring command data from the tape reader to internal storage without delaying the next incoming command and without interrupting the machine functions. The internal storage shall delay or store multiple blocks of data until required by the controlling devices.

3.4.10.20 Block-by-block read. The control shall have a block-by-block read function to provide for reading information one block at a time.

3.4.10.21 Tool length offsets. The control shall have a tool length offset feature to permit adjustments to the programmed tool length by means of manual data input. The tool length offset feature shall have a range of not less than the following:

a. Inch system +0.0002" to +1.0000".
b. Metric system +0.002 mm to +25.400 mm.

3.4.10.22 Cutter compensation. Cutter compensation shall be provided to compensate for cutter diameters differing from those utilized in programming. The cutter compensation shall have a range of not less than the following:

a. Inch system +0.0002" to +1.0000".
b. Metric system +0.002 mm to +25.400 mm.

3.4.10.23 Fixture compensation. A fixture compensation feature shall be provided to permit compensation for fixture position errors.

3.4.10.24 Reversal error compensation. Reversal error compensation shall be provided for each axis of motion to compensate for drive gear backlash

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when slide reversal occurs. A means shall be provided to input new compensation values when required.

3.4.10.25 Axis calibration. Error compensation shall be provided to correct repeatable errors determined to exist due to leadscrew and feedback device inaccuracies. A means shall be provided to input new error compensation values when required.

3.4.10.26 Additional features. Unless otherwise specified (see 6.2.1), the following additional numerical control features shall be provided:

a. Four quandrant programming.
b. Incremental and continuous jog for each axis.
c. Leading zero suppression or decimal point programming.
d. Inches Per Minute (ipm)/Millimeters Per Minute (mmpm), programming in all modes of operation.
e. Tape edit keylock.

3.4.10.27 Operator control panel. Unless otherwise specified (see 6.2.1), the control unit shall have a control panel which provides at least the following control functions: Emergency Stop; Cycle Start; Optional Stop; Mode Selection; Feed Hold; Feedrate Override; Spindle Speed Override; Jog; Jog Direction; and End-of-Block (EOB) Stop.

3.4.10.27.1 Pendant control. When specified (see 6.2.1), a pendant control shall be furnished in lieu of an operator control panel. The pendant control shall be connected to the machine to enable the operator to control machine functions while in a close observation position. Unless otherwise specified (see 6.2.1), the pendant control shall be capable of the same functions as stated for the operator control panel in paragraph 3.4.10.27.

3.4.10.28 Reference zero. The control unit shall have a reference zero to move each axis to a reference limit switch to establish synchronization and reset all registers to zero.

3.4.10.29 Set zero. The control unit shall have a “set zero” feature to permit the specified axis register to be reset to zero establishing present axis location as zero.

3.4.10.30 Program tryout. The control shall have a program tape tryout feature to permit new part programs to be read through the control without machine motions.

3.4.10.31 Operational software. Operational software shall be resident in the control and shall contain all logic necessary to effectively operate the machine tool. The software shall provide for meaningful diagnostics relating to the utilization of the controls, machine tool, part program, and operator’s actions.

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3.4.10.32 Programmable interface. A software program for accommodating the softwired interface between the numerical control and machine shall be provided as part of the executive program or as a separate program in the form of l-inch, eight-channel punched tape complying with EIA RS-244 or 8-inch flexible diskette as appropriate for the input media specified (see 6.2.1).

3.4.10.33 Maintenance diagnostics system. Maintenance diagnostics software shall be furnished in either the control as nonvolatile memory or in the form of l-inch, eight-channel punched tape complying with EIA RS-244 or 8-inch flexible disk ‘as appropriate for the input media specified (see 6.2.1) to provide for diagnostic checking of the machine control unit. The diagnostic-software shall-test, exercise, and display failures, at least to board level.

3.4.10.34 Peripheral equipment interface. Unless otherwise specified (see 6.2.1), the control shall be equipped with an interface complying with RS-232.

3.4.10.35 Axis jog. The control panel or pendant shall have manual controls for accomplishing both single and continuous, low and high speed jog movements of the slides for each controlled axis in both the plus and minus direction.

3.4.10.36 Auxiliary functions. Unless otherwise specified (see 6.2.1), the MCU shall respond to the following program commanded auxiliary functions: interpolar ion, fixed cycles, and dwell. Unless otherwise specified (see 6.2.1), miscellaneous functions shall include program stop, optional program stop, end-of-program, spindle clockwise (CW), spindle counterclockwise (CCW) , spindle off, coolant on-off, tool change, end-of-program (rewind), and program re-start. System preparatory and miscellaneous functions shall be coded in accordance with EIA RS-274.

3.4.10.37 Canned cycles. The following canned cycles shall be provided:

a. Drilling.
b. Boring.
c* Tapping.

3.4.10.38 Tape punch unit. When specified (see 6.2.1), the system shall have a tape punch unit for producing punched tapes from edited programs stored in data memory. The unit shall have a plug receptacle and circuitry that is compatible with the tape punch output of the system, punching both paper and mylar tapes of dimensions characteristics and code in accordance with 3,4.10.2.1. The punching speed of the unit shall be not less than 75 characters per second.

3.4.10.39 Postprocessor. When specified (see 6.2.1), the system shall be furnished with a postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall be

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provided for converting specified programming language (see 6.2.1) to the form and format required to operate the machine tool and control. The post-processor shall be written for use on the type computer specified (see 6.2.1).

3.4.11 Standard equipment. Unless otherwise specified (see 6.2.1) , the following equipment shall be furnished with the machine:

a. A load meter installed on the machine or MCU console that constantly registers the horsepower load (horsepower units or percent of load) on the spindle rotation drive motor.

b. One set of special wrenches, operating, and repair tools that are normally furnished by the manufacturer.

3.4.12 Optional equipment. The following optional equipment shall be furnished as specified (see 6.2.1):

a. A worklight, conveniently mounted to illuminate the work area, complete with protective shield and on-off switch.

b. Tools , tool holders, and adapters, as specified.

c. Other optional equipment, as specified.

3.5 Sizes and capacities. Unless otherwise specified (see 6.2.1), the sizes and capacities of the machining center shall meet the requirements of table II as specified for the machine size specified in the contract or order.

3.6 Performance. The machine and numerical control system combination shall be capable of performing as specified herein. In addition, the machine shall be capable of performing drilling, tapping, milling and boring to accomplish the capacities of table II while meeting the accuracies of table III.

3.7 Alignment. The machine and numerical control system shall meet the alignment and accuracy tolerances of table IV.

3.8 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be applied directly onto the item by environmentally protected decalcomania transfer, metal wrap-around tag, stencil, silk screen, or by any other method suitable for the use intended which shall be as permanent as the normal life expectancy of the item to which it is applied and capable of withstanding the environmental test and cleaning procedures specified for the item.

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3.8.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity
National Stock Number for lubricant

3.8.2 Nameplate. A nameplate shall be attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machining center is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.9 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of O to 99,999 hours in increments of 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.10 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.11 Workmanship. Workmanship of the machining center shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of type covered by this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.3, the tests in 4.4 through 4.4.6, and the inspection in 4.5. Failure of the item to pass any examination or test shall be cause for rejection.

4.3 Examination. The machine and equipment shall be examined for compliante with the requirements in 3.1 through 3.11.

4.4 Tests. The machine shall be subjected to the following tests. Al 1 instruments , tapes, materials, and tools required to perform and evaluate these tests shall be furnished by the supplier. The measuring instruments shall have evidence of calibration traceable to the National Bureau of Standards. The numerical control test tapes and computer printouts used shall be the property of the Government.

4.4.1 Operational test. The machine and its numerical control system shall be operated in accordance with the manufacturer's standard operating test procedure for warm-up and run-off checks. During the warm-up period, proper operation of all manual controls, motors, adjustment mechanisms and accessories shall be verified, After warm-up, the machine shall be cycled continuously under numerical tape control for a period of not less than 4 hours. This operation shall include tool changer, spindle speed and feed rate changes that include the highest, intermediate, and lowest settings of each range, rapid traverse of all slides, simultaneous movement of slides and automatic feed cycles as applicable. The numerical control system shall

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be further tested to verify proper operation of MDI, program edit, and system diagnostics. Should a malfunction occur, the fault shall be corrected and the operated test repeated from the beginning until a full 4 hours of running time is completed without failure or interruption. Failure to complete the operational test requirements after three attempts shall result in rejection of the machine.

4.4.2 Alignment and accuracy test. The machine and its numerical control system shall be tested for conformance with the alignment and accuracy requirements of table IV,

4.4.2.1 Axis positioning and overshoot tests. Starting at positions other than the extremes of axis travel, two identical movements of each controllable linear axis shall be programmed and executed in each direction. Programmed span shall be not less than 7 inches, and each digit of starting and end point shall be other than zero with respect to reference zero. The feed rate of each axis shall be 2 inches per minute. Absolute positioning and positioning repeatability error shall not exceed the permissible tolerances of table IV. Similar movements of each linear axis shall be prog rammed at various feed rates to test for slide overshoot at accelerated feed rates without programmed deceleration. One pass shall be programmed at feed rates nearest to 10, 20 and 1000 percent of maximum linear feed rate. Overshoot shall not exceed 0.003 inch at any feed rate up to 20 percent of the maximum. Maximum overshoot shall not exceed 0.005 inch. Stops shall be programmed appropriately for inspection of control positioning accuracy. Positioning tests may be part of the same tape program as 4.4.1.

4.4.3 Performance accuracy test. While using one or more test setups devised by the supplier, the machine shall be tested to verify its ability to perform machining operations under numerical control for meeting the accuracy tolerances of table 111. The test piece(s) may be, at the supplier's option, steel, cast iron or aluminum and the tool(s), spindle speed and feed rate for each test shall be selected by the supplier as the most suitable for the particular operation being performed. The test pieces(s) shall present a work surface normal to the tool point and approximately centered with reference to the work area in the xy plane. For the milling and contouring test, successive cuts may be made inside the outlines of previous cuts and roughing cuts may be made in the various cuts with excess metal removed as desired to make measurement convenient.

4.4.3.1 Tapping test. At programmed locations, pilot drill and tap four holes each with finished threads to accept 8-32 and 3/4-10 screws. Finished holes shall accept standard 8-32 or 3/4-10 screws freely and threads shall meet class 3 standards for internal threads as set forth in FED-STD-H28.

4.4.3.2 Boring test. Pilot drill, semi-finish bore, and finish bore not less than 10 holes in a selected pattern covering an area not less than 24 inches in diameter. Four of the holes shall be spaced on 18-inch centers. Different tools shall be used for semi-finish and finish boring of each hole shall be complete with necessary tool changes before the beginning boring

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operations of successive holes. Each address shall be to the fourth decimal place with digits other than zero. Finished hole size shall range from 1.125 to 2.750 inches. The resulting hole location and hole size errors shall not exceed the permissible tolerance of table III for boring.

4.4.3.3 Straight-line milling test. The straight line milling test shall be a programmed square or rectangle, not less than 6 inches per side with sides parallel to the travel along X and Y axes. Machines shall make a plunge cut along one longitudinal side cut. Plunge angle shall be not less than 2“. The resulting cuts shall be checked for dimensional accuracy, squareness, parallelism, flatness, angularity, and corner locations to verify the accuracy required in table III for straight-line milling.

4.4.3.4 Circle contouring test. A circle shall be programmed and milled within the outline of the square cut required in 4.4.3.3. The cut shall be made at constant Z-axis depth of one-half inch using the shortest programming sequence within the capability of the MCU. The resulting cut shall be checked for dimensional accuracy, roundness, surface finish, and angularity to verify the accuracy required in table III for circle contouring.

4.4.3.5 Straight-line contouring test. A square or an equilateral triangle shall be programmed and milled within the circle cut and described by 4.4.3.4 at constant depth of one-half inch. The reference line for one side of the square or triangle shall be 10 to 20 angular degrees canted to the X axis. The resulting cut shall be checked for dimensional accuracy, squareness, and angularity to verify the accuracy required in table III for straight-line contouring.

4.4.4 Maximum horsepower test. Using a workpiece of SAE 1020 steel, a straight cut of not less than 12 inches in length shall be made with a face mill or shell end mill. The diameter of the cutter and spindle speed shall be determined by the supplier to be suitable for making the cut with specified chip load and peripheral cutter speeds. The depth of the cut shall be as required to load the spindle motor to its maximum rated horsepower. Chip load shall be 0.010 +0.002 inch and peripheral cutter speed shall not exceed 300 feet per minute. The cutting action shall be smooth and even and the finished workpiece shall show no evidence of tool chatter. In addition, there shall be no evidence of overheating of the spindle drive motor or the applicable feed motor.

4.4.5 Test program verification. Unless otherwise specified (see 6.2.1), while using one or more test set-ups devised by the supplier, the machining test program (tape) shall be run on the machining center to verify its ability to receive and perform all program functions without malfunction over error. The test piece(s) shall be either steel, cast iron, or aluminum, while the tooling shall be selected by the supplier as the most suitable for the particular operation being performed.

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4.4.6 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control testing shall be tested for compliance with paragraph 3.3.8.

4.5 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging, packing, and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1) level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The computer numerically controlled machines covered by this specification are intended for use in any production shop where drilling, boring, tapping, reaming, and milling operations by numerical control of the machine are required.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in the procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c* Measurement system, US or SI, as specified (see 3.2).

d. System of graduation, US or SI, as specified (see 3.2.1).

e. Specify additional safety and health requirements (see 3.2.5).

f. Painting, if different (see 3.3.5).

g. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8 and 4.4.6).

h. Specify the number and configuration of T-slots, if different (see 3.4.3).

i, Sub-plate and desired characteristics, if required (see 3.4.3).

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j. Specify special characteristics for spindle noses and bores if required (see 3.4.4.1).

k. Electrical system, electrical supply, and voltage for initial wiring, if different (see 3.4.9).

l. Electric motor enclosures, if different (see 3.4.9.1).

m. Specify EIA-RS-494 Binary Cutter Location (BCL) data, if required (see 3.4.10).

n. Input media, if different from EIA RS-358 (see 3.4.10.2.1).

o. Specify EIA RS-244 in lieu of EIA RS-358 (see 3.4.10.2.1).

p. Feed and take-up spools, other than standard (see 3,4.10.2.1).

q. Data display, if different (see 3.4.10.3).

r. Specify memory type, if different (see 3.4.10.9).

s. Specify part program storage capacity, if different (see 3.4.10.9).

t. Specify the hours of standby battery power required, if different (see 3.4.10.10).

u. Specify additional NC features, if different (see 3.4.10.26).

v. Operator control panel functions, if different (see 3.4.10.27).

w. Pendant control panel, if required (see 3.4.10.27.1).

x. Pendant control panel functions, if different (see 3.4.10.27.1).

y. Specify peripheral equipment interface, if different (see 3.4.10.34).

z. Specify auxiliary and miscellaneous functions, if different (see 3.4.10.36).

aa. Tape punch unit, if required (see 3.4.10.38).

bb. Postprocessor, specify computer name and model, form and format (see 3.4.10.39).

cc. Specify standard equipment, if different (see 3.4.11).

dd. Optional equipment, if required; fully describe (see 3.4.12).

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ee. Specify sizes and capacities, if different (see 3.5).

ff. Lubrication chart, if different (see 3.8.1).

gg. Nameplate, if different (see 3.8.2).

hh. Hour meter, if not required (see 3.9).

ii. Quality conformance inspection, if different (see 4.2).

jj. Test program verification, if different (see 4.4.5).

kk. Preservation, packaging and marking, if different (see 5.1).

6.3 Contract data requirements. Required technical data (operating manual, parts list, wiring diagrams, and examination, original alignment and tolerance test reports and other instructions for operation and maintenance as identified on a DD Form 1664, Data Item Description) should be specified on a DD Form 1423, Contract Data Requirements List incorporated into the contract.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the requesting activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment.

6.5 Subject term (key word) listing.

Control, computer numerical
Column, double
Center, machining
Machine, milling
Spindle, vertical single
Tool, machine

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TABLE I. Diskette addressing.

Sector	*	Sector*	Byte	Use
0-25	0	All	All	Unused
26-29	1	1-4	0-1	File start sector (Binary)
			2-31	30 character ASCII program name to be defined by user
			32-63	Same as 0-31 above for next Part program (file)
			64-95	Same as 32-63 above for next part program
			96-127	Same as 32-63 above for next part program
		5-26	All	APT CL records for NC part programs
30-1897	2-73	All	All	APT CL records for NC part programs
1898-1975	74-76	All	All	Ususued

* This numbering corresponds to IBM System 3740 physical format.

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TABLE II. Sizes and capacities.

Characteristics	Requirements - Type I					
	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6
Spindle:						
Motor horsepower	20	20	20	25	30	30
Speeds, number	inf.	inf.	inf.	inf.	inf.	inf.
Speed range, rpm	45-2000	45-2000	45-2000	45-2000	45-2000	45-2000
Spindle assembly, Z-axis:						
Feed rate range, ipm	1-150	1-150	1-150	1-150	1-150	1-150
Rapid traverse, ipm	200	200	200	195	290	290
Travel, inches	15	15	15	15	25	25
Spindle carriage, Y-axis:						
Feed rate range, ipm	1-150	1-150	1-150	1-150	1-150	1-150
Rapid traverse, ipm	400	400	400	400	400	400
Travel, inches	50	62	62	78	98	115
Worktable, X-axis:						
Size, width x length, inches	38x60	38x70 (110)	45x80	50x90 (120) (180)	78x140 (180)	90x180
Feed rate range, ipm	1-150	1-150	1-150	1-150	1-150	1-150
Rapid traverse, ipm	400	400	400	400	400	400
Travel, inches	58	68 (108)	75	85 (115) (175)	130 (170)	170
Cross rail, W-axis:						
Travel, inches	38	38	38	45	50	58
Traverse speed, ipm	20	20	20	20	20	20
Capacity: (in SAE 1020 steel)						
Drill, dia., inches	2-1/2	2-1/2	2-1/2	2-1/2	3	3
Tap, dia., inches	2	2	2	2	2-1/2	2-1/2
Mill, cubic in. per min.	20	20	20	25	30	30

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TABLE II. Sizes and capacities - continued.

Characteristics	Requirements - Type II							
	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6	Size 7	Size 8
Spindle:								
Motor horsepower	10	25	25	25	25	25	25	25
Speeds, number	inf.	inf.	inf.	inf.	inf.	inf.	inf.	inf.
Speed range, rpm	40-3800	50-1600	40-3800	40-3800	40-3800	50-1600	40-3800	50-1600
Spindle assembly, Z-axis:								
Feed rate range, ipm	1-300	1-150	1-300	1-300	1-300	1-150	1-300	1-150
Rapid traverse, ipm	300	200	300	300	300	200	300	200
Travel, inches	18	30	18	18	18	30	18	30
Spindle carriage, Y-axis:								
Feed rate range, ipm	1-300	1-150	1-300	1-300	1-300	1-150	1-300	1-150
Rapid traverse, ipm	300	300	300	300	300	300	300	300
Travel, inches	30	78	48	48	48	78	66	110
Worktable, X-axis:								
Size, width x length, inches	30x50	38x110 (140)	41x64	42x98	48x32	50x140 (180)	60x146	70x140 (180)
Feed rate range, ipm	1-300	1-150	300	1-300	1-300	1-150	1-300	1-150
Rapid traverse, ipm	300	300	300	300	300	300	300	300
Travel, inches	50	118 (155)	60	96	30	140	146	140 (180)
Capacity: (in SAE 1020 steel)								
Drill, dia., inches	2	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2
Tap, dia., inches	1-1/2	2	2	2	2	2	2	2
Mill, cubic in. per min.	10	25	25	25	25	25	25	25

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TABLE III. Performance accuracy.

Machining position	Accuracy (Permissible tolerance)
Drilling:	
Depth of hole	±0.001"
Tapping:	
Threaded holes shall be class III fit in accordance with FED-STD-H28	
Boring and counterboring:	
Hole location, based on centerline distance of bored holes, 2 holes, 18 inches apart	±0.002"
Hole size, minimum spread, hole-to-hole	±0.002"
Depth of counterbore	±0.001"
Milling (straight-line):	
Squareness of sides	0.002"/FT
Parallelism of opposing sides	0.002"/FT
Flatness	±0.002"/3 FT
Angularity of ramp cuts	±0.002"/FT
Corner location, deviation from programmed points	
Contouring (circle):	
Roundness	
Diameters measured at 30 degree intervals	±0.002"
Radii measured at 30 degree intervals	±0.002"
Surface finish, maximum roughness 64 rms	
Angularity	
Difference between programmed and actual points of tangency of circles to straight surfaces	0.002"
Contouring (straight-line):	
Squareness of sides	0.002"/FT
Angularity of sides	0.002"/FT

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TABLE IV. Alignment and accuracy.

Component characteristic	Permissible tolerance
Worktable:	
Flatness of work mounting surface	0.001"/FT
Squareness of work mounting surface to spindle axis, in ay 6-inch sweep radius	0.002"/FT
Motion in the X axis, rise and fall with respect to the spindle	0.001"/FT
Motion in the Y axis, rise and fall with respect to the spindle	0.001"/FT
Parallelism of T-slots with X axis movement	0.0005"/FT
Squreness of T-slots with Y axis movement (slide travel squareness)	0.0005"/FT
Spindle:	
Axial runout	
Radial runout:	
1" from spindle nose	0.0005" TIR
6" from spindle nose	0.0015" TIR
Squareness of spindlke movement to the work mounting surface of the worktable per 6" of travel	0.0005"
Numerical control system:	
X and Y axes positioning and repeatability accuracy:	
Absolute positioning	1/ ±0.001"/FT
Positioning repeatability	±0.0005"
Z axis positioning and repeatability accuracy:	
Absolute positioning	±0.001"/FT
Positioning repeatability	±0.0005"
A axis (when furnished) positioning and repeatability accuracy:	
Absolute positioning	±10 sec./FT
Positioning repeatability	±5 sec.
Axis overshoot:	
20% of maximum feed rate	0.003"
Maximum feed rate	0.005"

NOTE: 1/ Not to exceed 0.025" in overall travel.

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

Preparing activity:
DLA - IP

Project (3408-0017)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-80269	Machining Centers, Vertical, Single Spindle, Double Column, Computer Numerically Controlled (CNC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*. _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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